THE UNIVERSITY OF WESTERN AUSTRALIA

CSSE

**SEMESTER 1, 2018 EXAMINATIONS**

**CITS4403**
**Computational Modelling**

FAMILY NAME: _____ GIVEN NAMES: _____

STUDENT ID: | | | | | | | | | SIGNATURE: _____

This Paper Contains: **15** pages **(including title page)**
Time allowed: **2:00 hours**

INSTRUCTIONS:

There are 23 questions. Marks for this paper total 120, Candidate must answer all questions.
Answers are to be written on this question paper in the spaces provided.

Calculators are not permitted.

**PLEASE NOTE**

Examination candidates may only bring authorised materials into the examination room. If a supervisor finds, during the examination, that you have unauthorised material, in whatever form, in the vicinity of your desk or on your person, whether in the examination room or the toilets or en route to/from the toilets, the matter will be reported to the head of school and disciplinary action will normally be taken against you. This action may result in your being deprived of any credit for this examination or even, in some cases, for the whole unit. This will apply regardless of whether the material has been used at the time it is found.

Therefore, any candidate who has brought any unauthorised material whatsoever into the examination room should declare it to the supervisor immediately. Candidates who are uncertain whether any material is authorised should ask the supervisor for clarification.

Supervisors Only Student left at:

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(Questions 1 to 22 carry 5 marks each)

1. Give pseudocode for an algorithm for determining if a graph is connected or not.

[5 marks]

2. Describe, using random graphs, an example of a surprisingly abrupt change in a property of a complex object when only gradual changes are made in parameters that govern the relationships between its components.

[5 marks]

3. Describe two different explanations for the Small World Phenomenon.

[5 marks]

4. Explain what the single-source shortest path problem is and describe briefly how to solve it.

[5 marks]

5. Explain the difference between a heavy-tailed distribution and one that follows a power law.

[5 marks]

6. Explain how to use preferential attachment to create random graphs with a distribution of degrees that obeys a power law.

[5 marks]

7. Consider the one-dimensional Wolfram-style Cellular Automaton called "Rule 18" operating on an infinite row of cells starting with only one cell on. Draw what the row looks like after 0, 1, 2 and 3 steps.

[5 marks]

8. Explain why Wolfram's demonstration of complex behaviour in cellular automata is considered more surprising than the complexity of chaotic systems.

[5 marks]

9. In the Game of Life, there is no initial condition that yields unbounded growth in the number of live cells. True or False? Explain your answer.

[5 marks]

10. Using the Game of Life explain realism versus instrumentalism.

[5 marks]

11. Explain briefly how reaction-diffusion could be responsible for animal patterns.

[5 marks]

12. Outline how to simulate percolation with a cellular automaton.

[5 marks]

13. Explain how the concept of self-organized criticality might suggest that large earthquakes are unpredictable.

[5 marks]

14. Explain how we can use a box counting algorithm to estimate fractal dimension.

[5 marks]

15. Use an example from the computational modelling course to explain and illustrate the idea of an emergent property.

[5 marks]

16. Explain how Schelling's model of segregation is a holistic model rather than a reductionist one.

[5 marks]

17. A very simple agent-based model of traffic evenly spread out along a road could easily seriously overestimate the flow. Explain how making a slightly more complicated, and slightly more realistic model of individual car behaviour, can suggest much more realistic and much lower traffic flows.

[5 marks]

18. Explain how it is possible to make a reasonably realistic model of a complex object like a flock of birds by just making a simple model of an individual bird.

[5 marks]

19. Outline some of mechanisms necessary for a process of evolution to allow agents to adapt to an environment.

[5 marks]

20. Explain how simulations of evolution can be used for optimisation, for example finding the arguments that give the maximum value of a function.

[5 marks]

21. Suppose that we have a population of agents capable of using a strategy to play the iterated Prisoner's Dilemma. Suppose we allow the strategies to evolve with fitness governed by success at the game. Describe some of the interesting population changes that may be observed.

[5 marks]

22. Explain the "problem of nice". Describe a solution.

[5 marks]

23. Briefly review a chapter of a popular non-fiction book that you have read recently covering a topic related to complexity science.

[10 marks]

END OF PAPER

Blank page for extra writing if needed